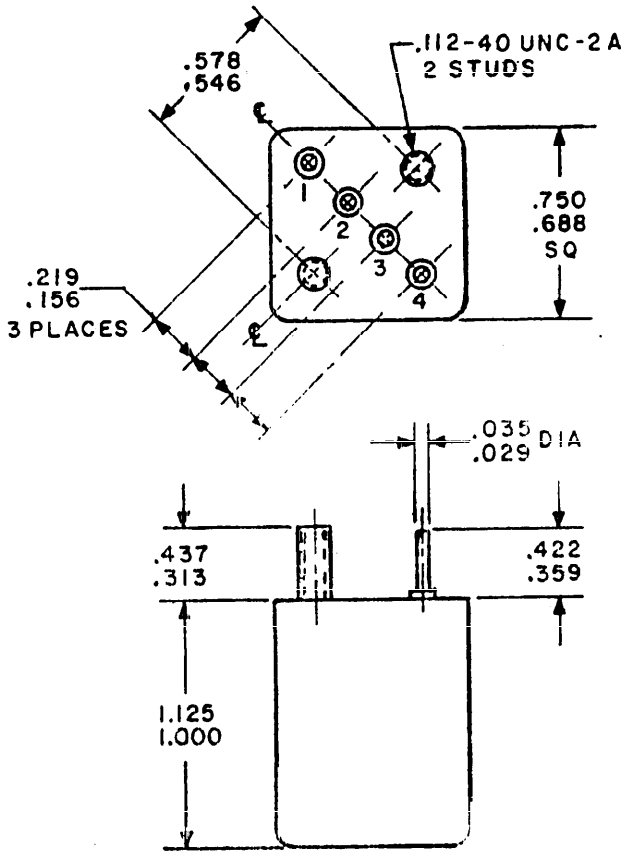
MILITARY SPECIFICATION SHEET

FILTER, BAND PASS, TYPE FR6RX22AF1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-18327.



INCHES	MM
.025	.64
.035	.89
.156	3.96
.219	5.56
.313	7.95
.359	9.12
.422	10.72
.437	11.10
.546	13.87
.578	14.68
.688	17.48
.750	19.05
1.000	25.40
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Terminal identification:
   Band Pass
   1 - 2 Input
   2 - 3 Output (10k)
   3 - 4 Output (30 Meg)
   Band Suppression
   4 - 1 Input (10k)
   1 - 3 Output (10K)

FIGURE 1. Case and mounting dimensions.

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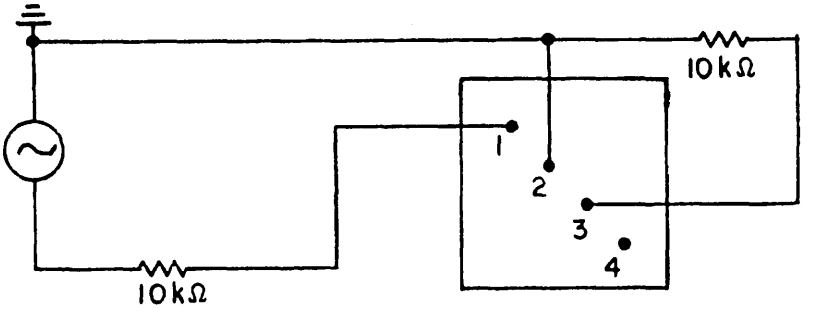


FIGURE 2. Connection for band pass, 10 dB I.L. MAX.

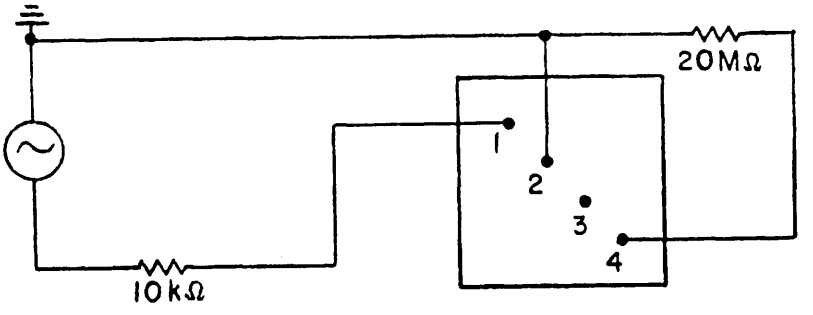


FIGURE 3. Connection for band pass 3 dB gain MIN.

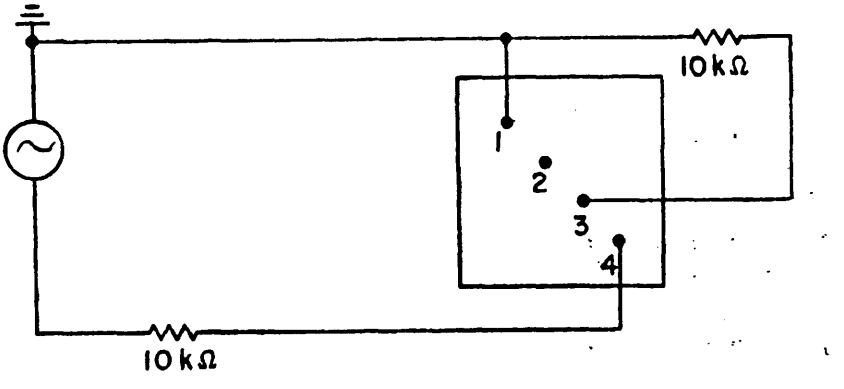


FIGURE 4. Connections for band suppression.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Metal.

Terminals: Solderable pin (see figure 1).

Weight: 1.5 ounces.

Terminal strength: Method 211, MIL-STD-202, test condition A - (Pull) force - 2 pounds.

Solderability: Method 208, MIL-STD-202. All terminations, of each filter, to be tested.

Solder dip dwell time: 5 ±1/2 seconds.

Examination of terminations: At 10X magnification.

Dielectric withstanding voltage:

At atmospheric pressure: Method 301, MIL-STD-202. 500 Vrms between each terminal and case.

At reduced pressure: Method 301, MIL-STD-202, and method 105, test condition B, MIL-STD-202. 450 Vrms between each terminal and case.

Insulation resistance: Method 302, test condition B, MIL-STD-202. Not less than 10,000 megohms when measured between each terminal and case.

Stability at temperature extremes:

1 hour at -55°C +0°C, -3°C.

1 hour at 105°C +3°C, -0°C.

Electrical characteristics: As specified herein.

Life: Method 108, test condition F, MIL-STD-202.

Test temperature: 105°C +10°C, -5°C.

Signal input level: .5 volt to source resistor.

DC operating voltage: Zero.

Measurements at 25°C after test:

Dielectric withstanding voltage: At atmospheric pressure.

Insulation resistance: Not less than 1,000 megohms between each terminal and case.

Electrical characteristics: As specified herein.

Temperature rise: Not applicable.

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electrical characteristics: These filters are designed for both band pass and band suppression applications.

Band pass: See table I, figure 2, and figure 3.

Impedance: The source impedance shall be 10 kilohms. The load impedance shall be 10 kilohms with 2-3 as output and 20 megohms with 2-4 as the output.

Insertion loss: The insertion loss at the reference frequency shall be 10 dB maximum with 2-3 as the output and 3 dB minimum gain with 2-4 as the output.

Band suppression: For band suppression connect the filter as shown on figure 4. The electrical characteristics shall be as shown in table II and as specified herein.

Impedance: The source and load impedances shall be 10 kilohms.

Insertion loss: The insertion loss at the reference frequency shall be 10 dB maximum.

TABLE I. Band pass electrical characteristics.

Dash number	Reference frequency (Hz)	Frequency (Hz) Discrimination			
		2 dB maximum		35 dB minimum	
				≤	≥
001	405	393 to	417	202.5	810
002	400	388	412	200	800
003	440	427	453	220	880
004	500	485	515	250	1,000
005	600	582	618	300	1,200
006	800	776	824	400	1,600
007	1,000	970	1,030	500	2,000
008	1,200	1,164	1,236	600	2,400
009	1,500	1,455	1,545	750	3,000
010	1,600	1,552	1,648	800	3,200
011	2,000	1,940	2,060	1,000	4,000
012	2,500	2,425	2,575	1,250	5,000
013	3,000	2,910	3,090	1,500	6,000
014	3,200	3,104	3,296	1,600	6,400
015	4,000	3,880	4,120	2,000	8,000
016	4,800	4,656	4,944	2,400	9,600
017	5,000	4,850	5,150	2,500	10,000
018	6,000	5,820	6,180	3,000	12,000
019	8,000	7,760	8,240	4,000	16,000
020	10,000	9,700	10,300	5,000	20,000
021	20,000	19,400	20,600	10,000	40,000

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TABLE II. Band suppression electrical characteristics.

Dash number	Reference frequency (Hz)	Frequency (Hz) Discrimination			
		25 dB maximum		2 dB minimum	
		≥	≤	≤	≥
001	40	393	417	202.5	810
002	40	388	412	200	800
003	44	427	453	220	880
004	50	485	515	250	1,000
005	60	582	618	300	1,200
006	80	776	824	400	1,600
007	100	970	1,030	500	2,000
008	120	1,164	1,236	600	2,400
009	150	1,455	1,545	750	3,000
010	160	1,552	1,648	800	3,200
011	200	1,940	2,060	1,000	4,000
012	260	2,425	2,575	1,250	5,000
013	300	2,910	3,090	1,500	6,000
014	320	3,104	3,296	1,600	6,400
015	400	3,880	4,120	2,000	8,000
016	460	4,656	4,944	2,400	9,600
017	500	4,850	5,150	2,500	10,000
018	600	5,820	6,180	3,000	12,000
019	800	7,760	8,240	4,000	16,000
020	1,000	9,700	10,300	5,000	20,000
021	2,000	19,400	20,600	10,000	40,000

Vibration: Method 204, test condition B, MIL-STD-202. Electrical load not applicable.

Measurements at 25°C after test:

Dielectric withstanding voltage: At atmospheric pressure.

Electrical characteristics: As specified herein.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Measurements at 25°C after test:

Dielectric withstanding voltage: At atmospheric pressure.

Electrical characteristics: As specified herein.

Thermal shock: Method 107, test condition A, MIL-STD-202, except step 3 shall be 105°C.

Exposure time: 1 hour.

Measurements at 25°C after test:

Dielectric withstanding voltage: At atmospheric pressure.

Insulation resistance: Not less than 10,000 megohms between each terminal and case.

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Immersion: Method 104, test condition B, MIL-STD-202. Immersed for 15 minutes in saturated solution of sodium chloride and water.

Measurements at 25°C after test:

Dielectric withstanding voltage: At atmospheric pressure.

Insulation resistance: Not less than 10,000 megohms between each terminal and case.

Moisture resistance: Method 106, MIL-STD-202.

Measurements at 25°C after test:

Dielectric withstanding voltage: At atmospheric pressure.

Insulation resistance: Not less than 10,000 megohms between each terminal and case.

Part number: M18327/024 -(dash number from table I).

QUALITY ASSURANCE PROVISIONS

Qualification of filters based on similarity to qualified filters: Qualification by similarity may be granted to all parts covered in this specification based on the qualification of one part number. Qualification by similarity shall be restricted to those parts which meet the criteria for a similar filter as specified in MIL-F-18327. The contractor shall provide certification that any parts qualified on the basis of similarity meet the full requirements of this specification and MIL-F-18327.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5915-0242-2)

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